

Matrix functions and stability

# Algebraicity of joint spectral radii from rational input

**Difficulty:** extreme **Importance:** interesting to the community

**Status:** Open

**Rating rationale:** Extreme because exact arithmetic character of rational-input growth rates remains a longstanding barrier; community impact concerns representability and certification of spectral computations.

## Context and notation

The joint spectral radius of a nonempty compact set  $\mathcal{M} \subset \mathbb{C}^{d \times d}$  is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written  $\rho(A)$ .

## Problem statement

Is it true that, for every  $d \geq 1$  and every finite nonempty  $\mathcal{M} \subset \mathbb{Q}^{d \times d}$ , there exists a nonzero polynomial  $p \in \mathbb{Z}[t]$  such that

$$p(\hat{\rho}(\mathcal{M})) = 0?$$

The claim is about the value being algebraic; it does not assert a uniform procedure for finding its minimal polynomial.

## Reference and status evidence

Jungers, *The Joint Spectral Radius: Theory and Applications*, §4.4, Open Question 7, printed p. 75 of the author manuscript. Searches combining “joint spectral radius”, “rational matrices”, “algebraic number”, “algebraicity”, and “transcendental” located no proof or rational-input counterexample. The latest explicit formulation located remains the book.

## Scope

Both signs are permitted. A finite-product formula would imply algebraicity for its particular input, but this question requires no such formula. It concerns exact representation of a matrix computation’s output, separately from the geometry of the set of all stable inputs.

## Audit — 2026-09-10

Rechecked the indexed text of Jungers, Open Question 7 and searched rational-input algebraicity and transcendence. No proof or rational counterexample was located. Failure of semialgebraic dependence on input data does not imply a transcendental value for a rational input; recent explicit open-status evidence remains unavailable.